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Document ID: PAY-011 | Classification: Internal Use

Refund and chargeback processing: initiation, status tracking, dispute resolution, and reason codes.

Error 5003 indicates gateway timeout. Typical causes: network issues, gateway overload, DNS failure.

Payment database management: transaction logs, audit trails, backup procedures, and data retention policies.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

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Payment processor integration: Stripe, Adyen, Braintree API documentation and migration guides.

Connection pooling for payment DB: set max_connections=50, connection_timeout=30s.

Payment gateway configuration for EU region: PSD2 compliance, SCA requirements, and GDPR data handling.

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Financial compliance: PCI-DSS Level 1, SOC 2, ISO 27001 requirements for payment systems.

For INR to USD discrepancy: verify FX rate source, conversion timestamp, and rounding rules.

Payment user authentication: 3DS2 flows, biometric verification, and risk-based authentication.

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Transaction reconciliation: daily settlement, mismatch resolution, and reconciliation report generation.

Chargeback reason code 10.4: customer disputes transaction. Gather proof of delivery and authorization.

Payment transaction logging: structured logs, trace IDs, and observability for incident investigation.

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Payment fraud detection: velocity checks, device fingerprinting, and ML-based risk scoring.

Payment API rate limit: 100 requests/second. Use exponential backoff for 429 responses.

Payment service deployment: blue-green releases, canary deployments, and rollback procedures.

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Merchant account management: onboarding, KYC verification, and settlement schedule configuration.

PCI-DSS: Never log full card numbers. Mask to last 4 digits. Secure log storage required.

Billing and invoicing: subscription management, proration, tax calculation, and invoice generation.

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